آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Computer Science Annual Examinations 2025

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Computer Science. It contains comments on candidates' responses to the 2025 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding, and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs, which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used. It is imperative to refer to the command word guide available on AKU-EB website for understanding the expectations of the command word.

General Observations

Most candidates succeeded in constructing good responses, particularly on the topics of system development life cycle (SDLC), introduction to C++ programming, and control structures. However, teachers should focus on the following areas and provide more drills and practice to foster a solid understanding:

- Operating Systems
- Object Oriented Programming (OOP)
- File Handling
- Operations in Programming
- Switch Statement
- Functions
- Array and Strings

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

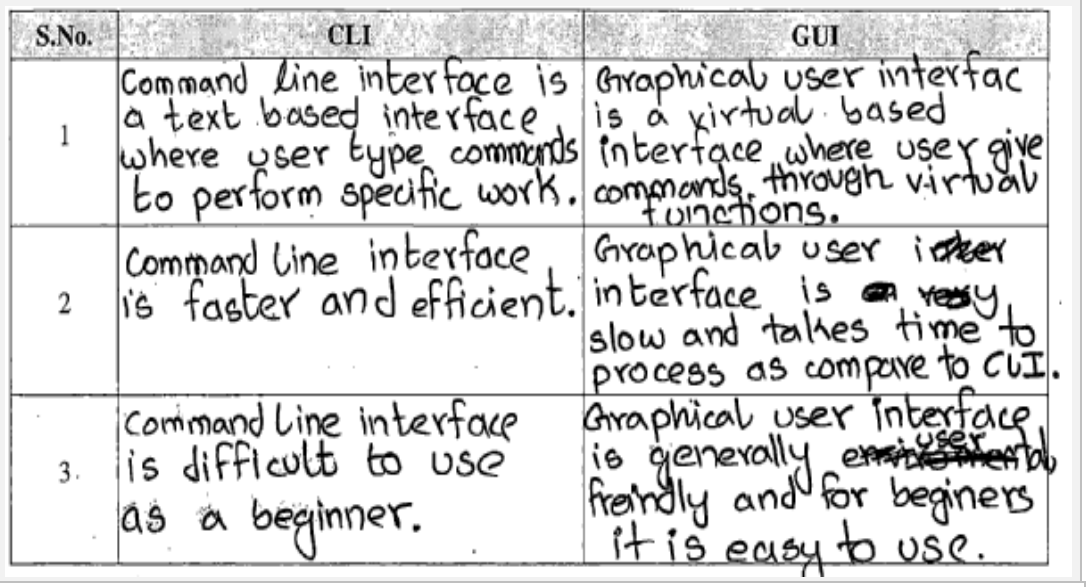
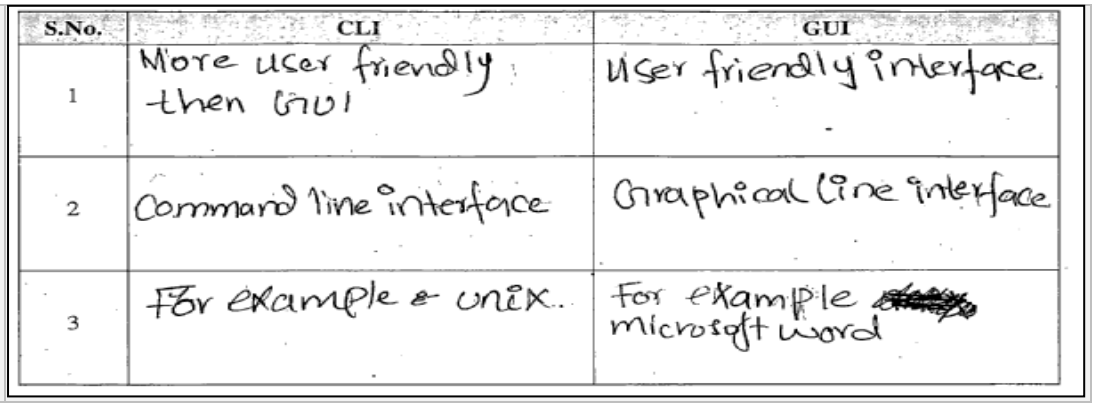
Question No. 1	
Question Text	Write any THREE differences between Command Line Interface (CLI) and Graphical User Interface (GUI).
SLO No.	9.1.3
SLO Text	Differentiate between command line interface and graphical user interface of an operating system.
Max Marks	3
Cognitive Level	Understand
Checking Hints	1 mark for writing each difference (any THREE required). Note: If the candidate mentions only one side, no marks will be awarded.
Overall Performance	The question was satisfactorily attempted by the candidates. Some candidates correctly stated three relevant differences between CLI and GUI, showing a basic conceptual understanding. However, a portion of candidates provided vague, irrelevant, or incomplete differences.
Description of Better Responses	*In better responses*, candidates accurately compared the CLI and GUI across valid parameters, such as GUI is easy to use compared to CLI, user interaction through commands in CLI, while through visual display in GUI, and CLI is faster in as compared to GUI. Their answers were clearly structured as point-wise comparisons.
Image of Better Response	
Description of Weaker Responses	*In weaker responses*, candidates showed confusion and provided inaccurate comparisons. For example, some mentioned that the command line interface is more user-friendly than the graphical user interface. They only wrote the terms 'command line interface' and 'graphical user interface without further clarification. Although a few examples, such as UNIX and Microsoft Word, were included, they were not properly linked to the differences in usability or interaction.

Image of Weaker Response	

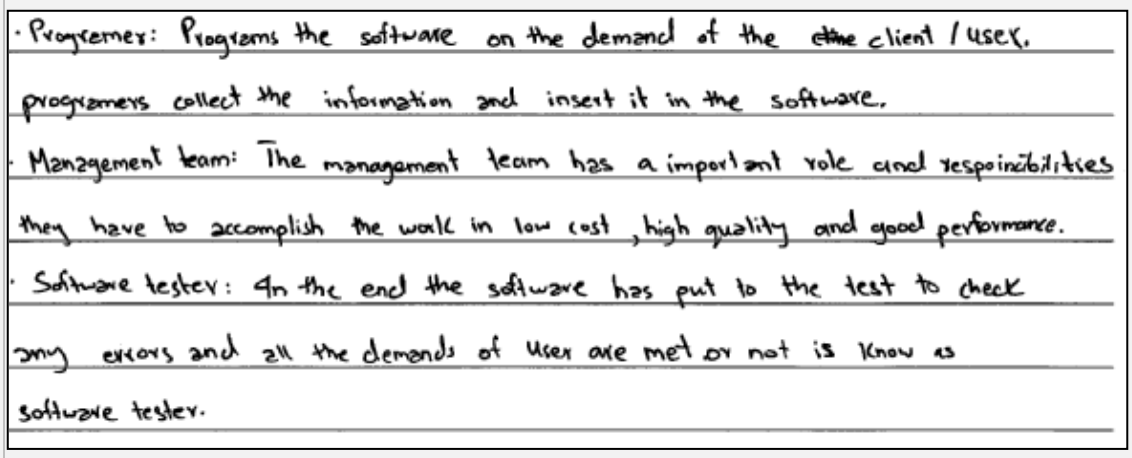
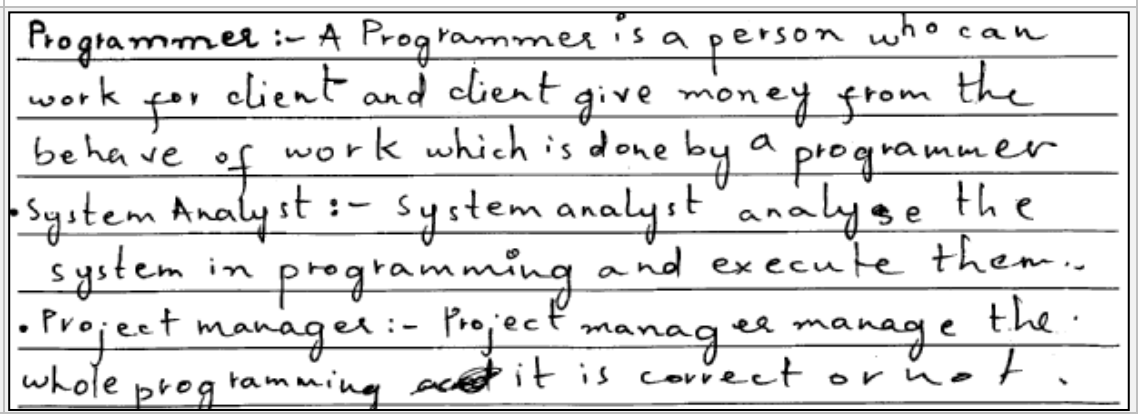
S.No.	CLI	GUI
1	More user friendly then GUI	User friendly interface.
2	Command line interface	Graphical line interface
3	For example e unix..	For example microsoft word

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy**Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Teachers are suggested to incorporate practical activities, such as live demonstrations, guided exercises, and peer comparisons, so that students can directly explore the differences between CLI and GUI. This hands-on approach will help learners accurately link examples like UNIX and Microsoft Word to usability, speed, and interaction, improving understanding and reducing misconceptions.

Question No. 2	
Question Text	Describe the responsibilities of any THREE of the following personnel in System Development Life Cycle (SDLC). • Management team • System analyst • Project manager • Programmer • Software tester
SLO No.	10.1.4
SLO Text	Describe the responsibilities of the following personnel: * a. management team b. system analyst c. project manager d. programmer e. software tester f. customer.
Max Marks	3
Cognitive Level	Understand
Checking	1 mark for describing each personnel (any THREE required).

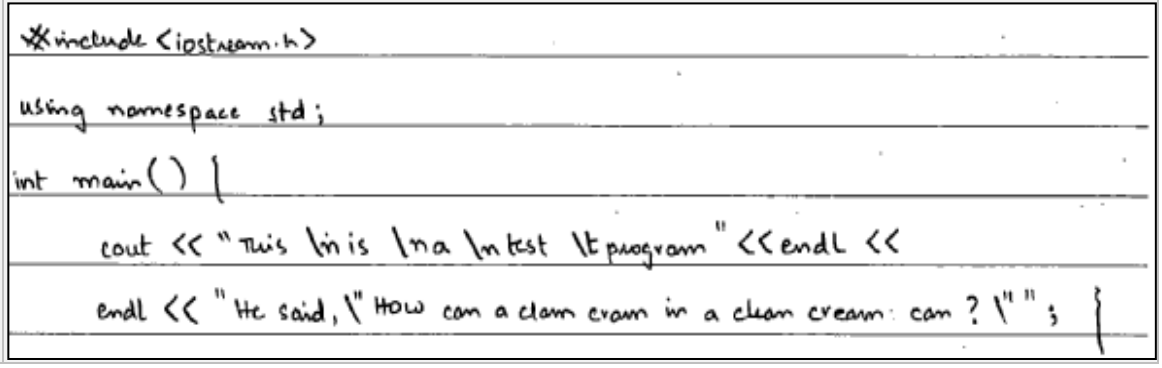
Hints	
Overall Performance	The question was well attempted by most candidates. Many were able to describe the responsibilities of three SDLC personnel clearly and concisely. However, a few candidates struggled to differentiate between the roles or used generic terms without linking them to SDLC.
Description of Better Responses	*In better responses*, candidates correctly identified the selected personnel and described their specific responsibilities within the SDLC. For example, programmers were described as writing code based on system requirements; management teams were linked to resource allocation and goal setting; software testers were accurately connected with error checking and validation; system analysts examine business needs and design system solutions to meet them, and project managers plan, organise, and oversee projects to ensure timely and successful completion.
Image of Better Response	
Description of Weaker Responses	*In weaker responses*, candidates wrote vague definitions that were not contextualised within the SDLC. Some responses have mixed roles. For example, giving tester duties to a programmer. They wrote programmer only works for the client and takes money; similarly, the project manager manages the programming. A few only listed the names without explaining responsibilities or wrote responsibilities unrelated to software development.
Image of Weaker Response	

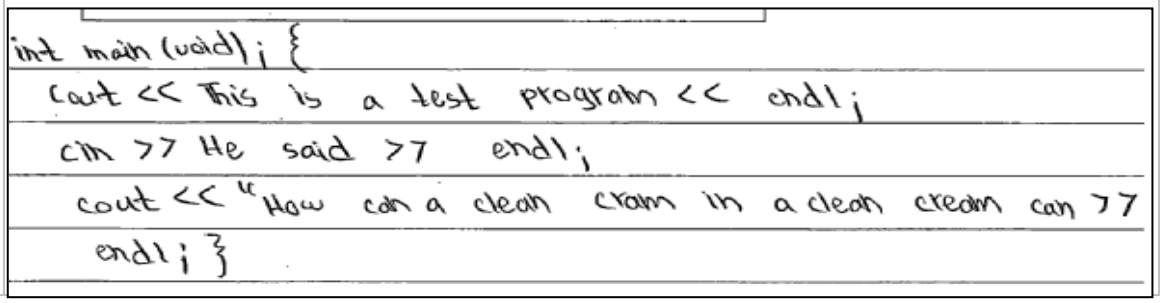
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy**Used for this SLO	Assessment Strategies
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cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion: Teachers are suggested to design activity-based exercises, such as role-play or simulation of a software development project, where candidates take on specific roles like programmer, tester, or project manager. They should perform tasks and document their responsibilities within each SDLC phase.

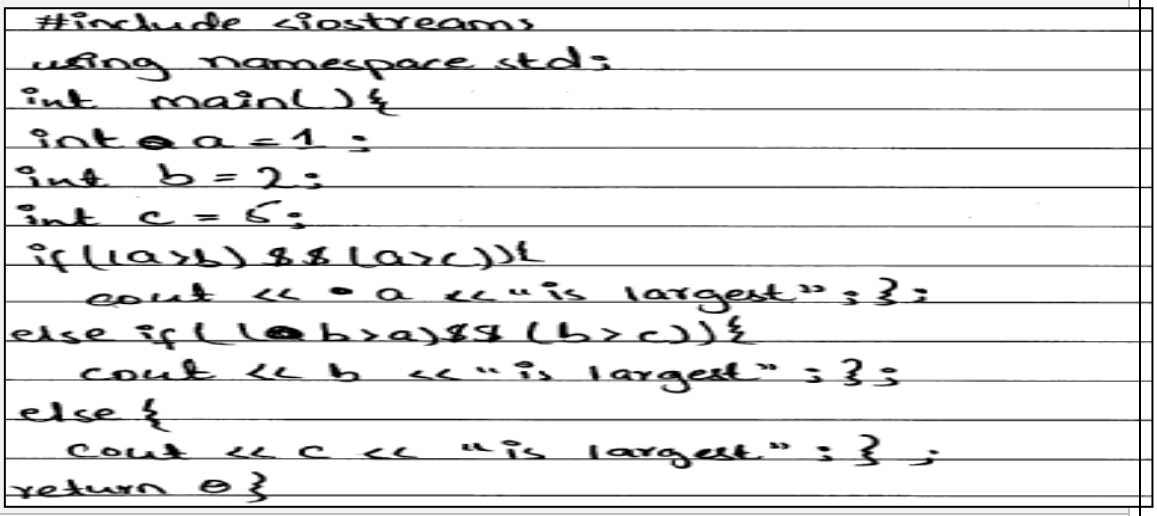
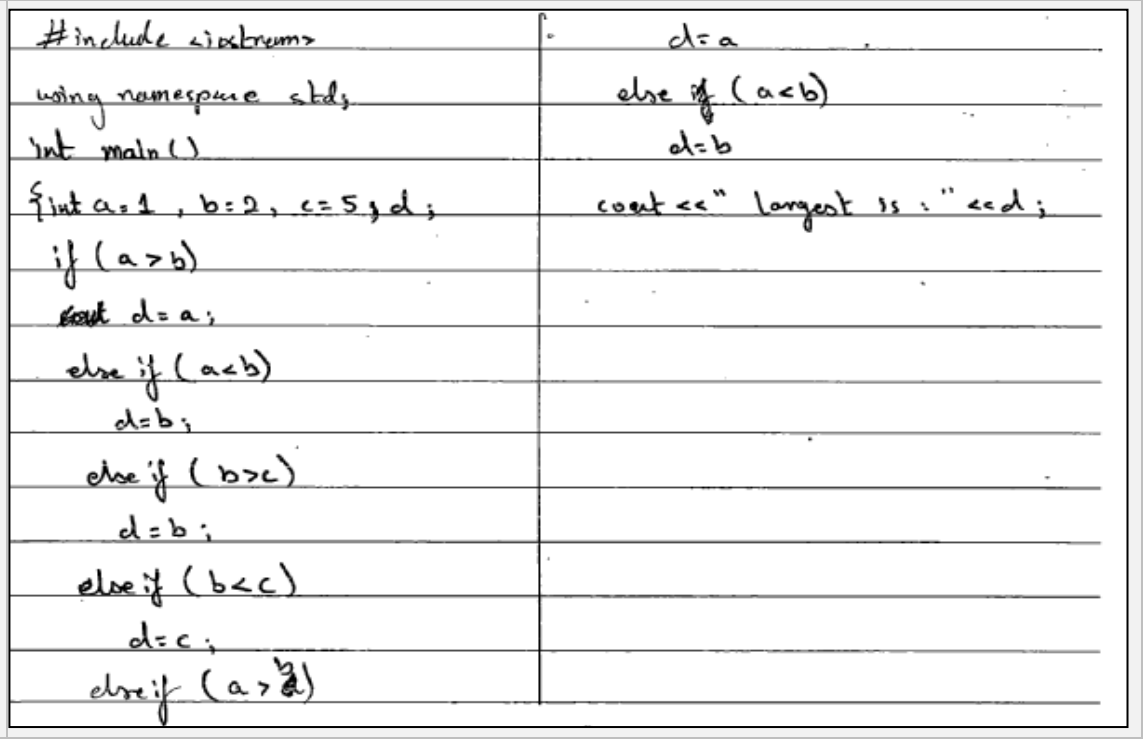
Question No. 3(i)	
Question Text	Write a program to display the following output. This is a test program He said, "How can a clam cram in a clean cream can?"
SLO No.	11.3.4
SLO Text	Write a program to print text using the following escape sequences: * a. alert – \a b. backspace – \b c. newline – \n d. carriage return – \r e. tab – \t f. display backslash – \\ g. display single quotation marks – \'.
Max Marks	2
Cognitive Level	Application
Checking Hints	1 mark for writing cout << "This\nis\na\ntest\tprogram\n\n"; 1 mark for writing cout << "He said,\"How can a clam cram in a clean cream can?\"\n"; 1 mark will be given overall if there is a syntax mistake, but the student has at least use escape sequence. No mark deduction for using /n instead of \n.
Overall Performance	Very few candidates demonstrated a clear understanding of output formatting concepts in C++. They effectively used escape sequences or appropriate alternatives to produce the required output. However, most of the responses contained small but critical syntax errors or failed to fully meet formatting expectations.
Description of Better Responses	*In better responses*, candidates wrote a well-structured C++ program that applied \n, \t, and \" accurately. Some candidates used endl in place of \n or applied setw( ) as an alternative to \t with equally correct output. These candidates showed a good grasp of both the syntax and the formatting logic.
Image of Better Response	

Description of Weaker Responses	*In weaker responses*, candidates reflected gaps in understanding escape sequences or their placement. Several candidates omitted necessary quotation marks or inserted escape characters incorrectly, leading to distorted output. A few candidates attempted the logic well but lost marks due to syntax issues, such as missing semicolons or misuse of cout.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy**Used for this SLO	Assessment Strategies
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Any Additional Suggestion: Teachers are suggested to organise hands-on coding exercises where candidates practice writing strings with various escape sequences, including placing them correctly within quotation marks and combining them with cout statements. Immediate feedback on syntax errors like missing semicolons or misplaced escape characters will help candidates solidify correct usage and prevent common mistakes.		

Question No. 3(ii)	
Question Text	Write a program to find the largest number among 1, 2 and 5.
SLO No.	11.4.8
SLO Text	Write a program using relational operators.
Max Marks	3
Cognitive Level	Application
Checking Hints	1 mark for declaring and initialising variables for given values. 1 mark for applying conditions (1st Comparison). 1 mark for applying conditions (2nd Comparison).
Overall Performance	Candidates' performance on this question was average. While few candidates handled the conditional logic effectively and produced correct output, many others struggled with syntax or only compared two numbers instead of all three. The logic for determining the maximum value was not consistently applied across all responses.
Description of Better Responses	*In better responses*, candidates wrote well-written C++ programs using if-else or nested if statements. For example, 1st number with the other two numbers, similarly, 2nd number, and 3rd number with the other two numbers. Some candidates even used the logical and relational operators, demonstrating confidence with alternative approaches.

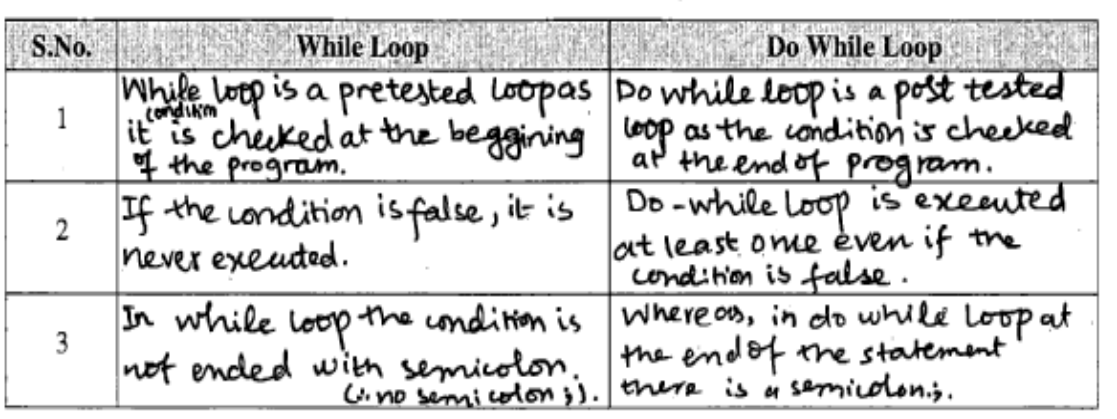
Image of Better Response	
Description of Weaker Responses	*In weaker responses*, candidates used incorrect comparison operators, leading to flawed logic, and failed to include the third variable in their comparisons or used improper syntax for statements such as printf, if, and else-if. In several cases, it was evident that candidates lacked a clear understanding of how to apply conditional statements correctly, indicating a need for further reinforcement of fundamental programming concepts.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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• Review past paper questions on the concept • Utilise the resource guide for additional materials	• Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion: Teachers are suggested to guide candidates in the step-by-step construction of conditions and encourage them to dry-run their logic before coding. Classroom activities such as live coding with random values can help students better understand conditional comparisons.

Question No. 4	
Question Text	Write any THREE differences between while and do while loop.
SLO No.	12.2.5
SLO Text	Differentiate between: * a. for and while loop b. while and do while loop.
Max Marks	3
Cognitive Level	Understand
Checking Hints	1 mark for each correct difference (any THREE Required). (Note: if the candidate mentions only one side, no marks will be awarded.
Overall Performance	The responses to this question reflected a low than moderate understanding of loop structures. While some candidates identified key differences, such as condition checking, position, and guaranteed execution, others presented incomplete or misinterpreted comparisons. The confusion between the syntax and flow of control was a common issue among weak responses.
Description of Better Responses	*In better responses,* candidates clearly wrote the correct difference between the while loop and the do-while loop. For example, a while loop was a pre-test loop, and a do-while loop was a post-test loop. If the condition was false in the while loop at first, it could not be executed, whereas the do-while loop executed at least once. The while loop was called an entry-controlled loop, while the do-while was an exit-controlled loop; no terminator was placed at the end of the loop in a while loop, and a terminator was required at the end of the loop in a do-while loop. Syntax and flow diagrams were sometimes included to support the explanation.
Image of Better Response	
Description of Weaker Responses	*In weaker responses*, candidates mentioned superficial or incorrect points, such as claiming one loop is faster or "used more commonly." Some candidates confused the placement of the condition, while others repeated the same point in different words or left one or more differences incomplete.

S.No.	While Loop	Do While Loop
1	While loop is a pretested loop as it (condition) is checked at the beggining of the program.	Do while loop is a post tested loop as the condition is checked at the end of program.
2	If the condition is false, it is never executed.	Do-while loop is executed at least once even if the condition is false.
3	In while loop the condition is not ended with semicolon. (i.e no semicolon ;).	Whereas, in do while loop at the end of the statement there is a semicolon ;.

Image of Weaker Response

S.No.	While Loop	Do While Loop
1	It is used when the code has to be run multiple times.	It is used is incase the codes has to be run multiple time.
2	It gives an answer after completing the code run.	It is able to give answer before running of code multiple time.
3	It is a slower process takes time to show the output	It is a fast process and takes less time to show the output

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion: Teachers are suggested to provide ample practice with converting one loop type to another and encourage candidates to trace the loop execution step-by-step. Group discussions using example code snippets can also help reinforce structural and logical distinctions between loop types.

Question No. 5

Question Text

Write a C++ program to illustrate single inheritance. The program should include the following:

- A base class named Company with a variable named CompanyName that stores the string "Softech Innovators Ltd."
- A derived class named Employee that inherits from the Company. It should also store the string "Ahmed Raza" in the variable EmployeeName, "Engineering" in the variable Department, and "Software Engineer" in the variable JobTitle.
- Create an object of the Employee class named empObj to display the company name, employee name, department, and job title.
- The output should be the same as shown below:

```
Company Name: Softech Innovators Ltd.
Employee Name: Ahmed Raza
Department: Engineering
Job Title: Software Engineer
```

SLO No.	16.3.8
SLO Text	Illustrate types of inheritance in C++ programming using different classes.
Max Marks	4
Cognitive Level	Application
Checking Hints	1 mark for storing string data in base class. 1 mark for storing string data in derived class. 1 mark for creating object of Employee class. 1 mark for display output.
Overall Performance	This question challenged candidates conceptually and syntactically. While several candidates were able to set up the class structure correctly, only a few successfully implemented inheritance, object creation, and output. Many responses indicated a surface-level understanding of inheritance without full execution of the task requirements.
Description of Better Responses	*In better responses,* candidates wrote a well-structured code in C++, a base class named Company was defined with a string variable CompanyName initialised to "Softech Innovators Ltd." A derived class named Employee was then created, inheriting publicly from the Company class. Within the Employee class, three additional string variables were declared and initialised: EmployeeName as "Ahmed Raza," Department as "Engineering," and JobTitle as "Software Engineer." In the main( ) function, an object named empObj of the Employee class was created and used to display the company name, employee name, department, and job title. The program produced the exact output shown in the image, demonstrating a clear understanding of basic inheritance, proper use of object handling, and correct implementation of C++ syntax.

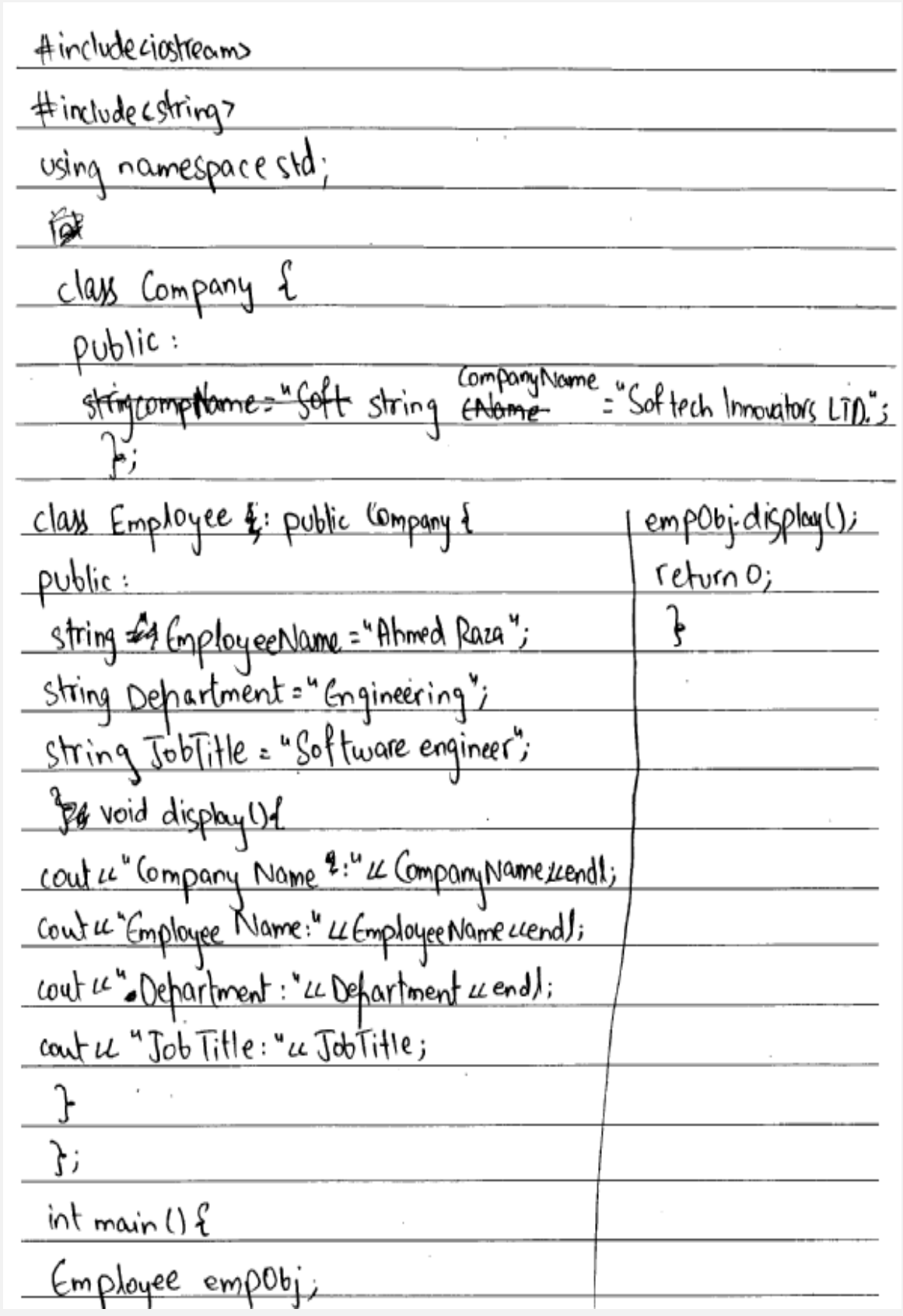
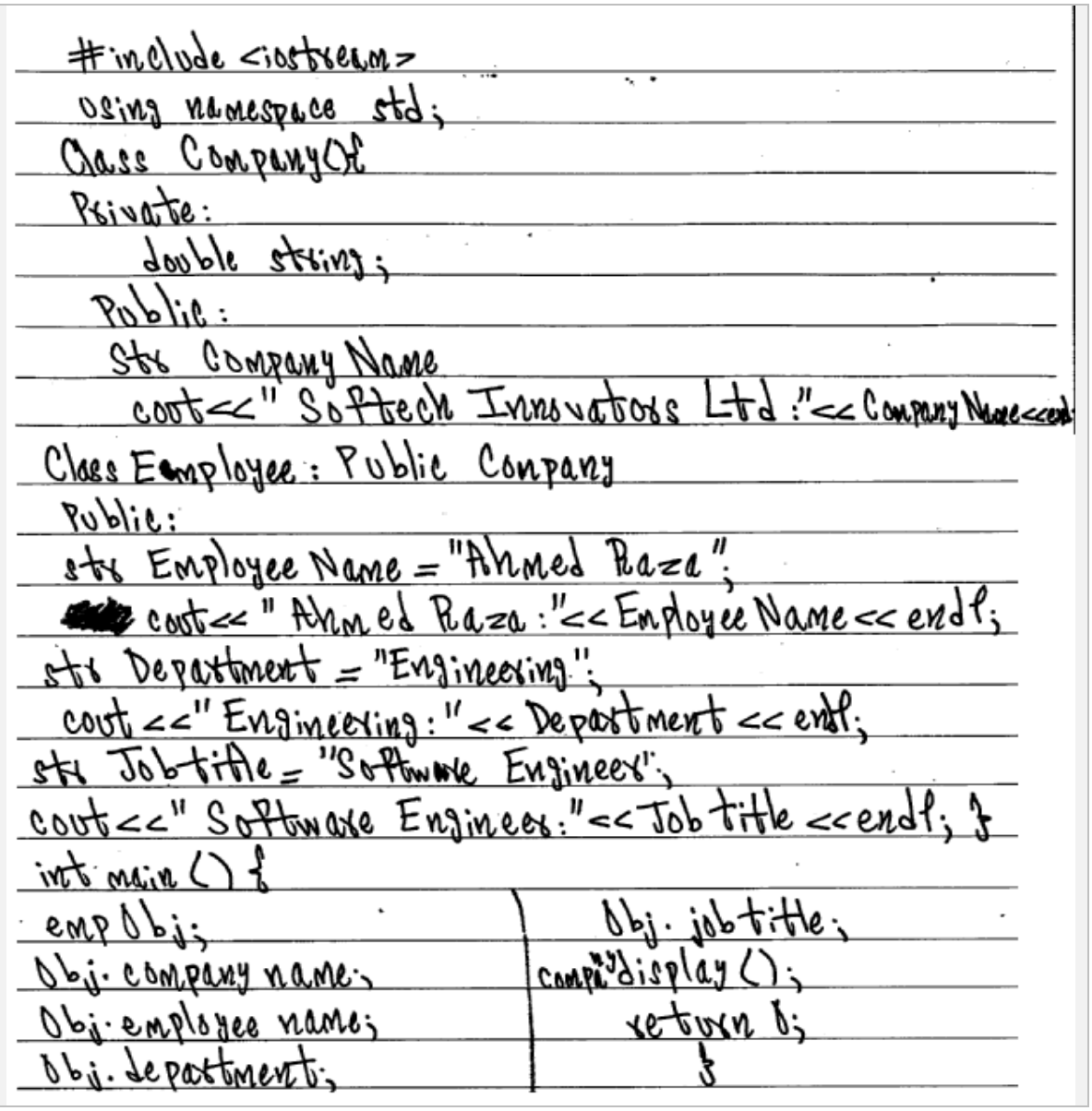
Image of Better Response	
Description of Weaker Responses	*In weaker responses*, candidates showed several common issues, such as some candidates wrote two independent classes instead of using inheritance, others failed to declare variables properly or skipped value assignment, a few had corrected class structure but created objects incorrectly, or didn't display the values. Syntax errors such as missing access specifiers or incorrect cout statements were frequent.

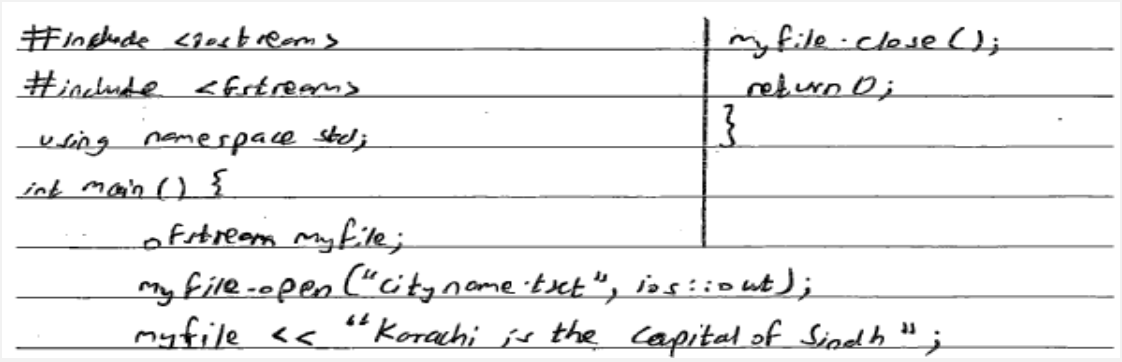
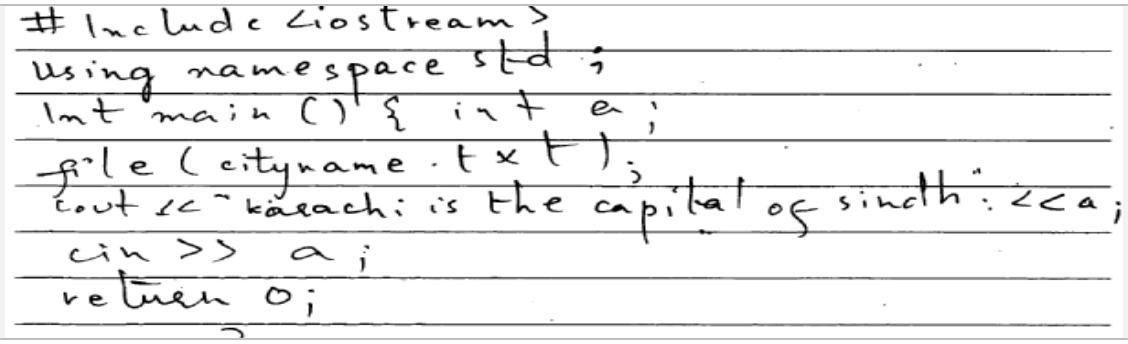
Image of Weaker Response	

```
#include <iostream>
using namespace std;
Class CompanyOf
Private:
    double string;
Public:
  Str Company Name
    cout<<" Softech Innovations Ltd :"<< Company Name<<end
Class Employee : Public Company
Public:
str Employee Name = "Ahmed Raza";
    cout<<" Ahmed Raza :"<< Employee Name << endl;
str Department = "Engineering";
 cout <<" Engineering :" << Department << endl;
str Jobtitle = "Software Engineer";
cout<<" Software Engineer :"<< Job title << endl; }
int main () {
 emp Obj;                          Obj. job title;
Obj. company name;                 compy display ();
Obj. employee name;                  return 0;
Obj. department;                   }
```

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion: Teachers are suggested to ensure that the candidates understand not just how to use inheritance, but why it's used. Encourage writing small-scale inheritance examples in the lab and illustrating object flow through diagrams. Have candidates explain inheritance in their own words to solidify understanding.

Question No. 6	
Question Text	Write a C++ program that • creates a text file named as cityname.txt • writes following line in the file. “Karachi is the capital of Sindh” • closes the file.
SLO No.	17.1.3
SLO Text	Write programs to show the file opening and closing.
Max Marks	3
Cognitive Level	Application
Checking Hints	1 mark for fstream pre-processing directive & object creation. 1 mark for creating cityname.txt file. 1 mark for writing the text in the file.
Overall Performance	The responses to this question were not good at all. Only very few candidates demonstrated an accurate understanding of file handling in C++, while many struggled with the syntax and flow. The question required familiarity with ofstream, which several candidates either misused or omitted altogether. This indicated a need for more hands-on practice with file operations.
Description of Better Responses	*In better responses*, candidates wrote a well-structured program with the use of #include<fstream>, proper declaration of ofstream, and the use of open( ), << operator to write, and close( ) to end the file session. The file was named appropriately, and the text was written as required.
Image of Better Response	
Description of Weaker Responses	*In weaker responses*, some candidates confused file creation with normal screen output, some forgot to include the fstream library, or incorrectly declared the file stream, and a few only wrote the main( ) structure without any valid file-handling code. Common syntax mistakes included not calling .open( ) or .close( ), or misusing the insertion operator (<<) were found commonly.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

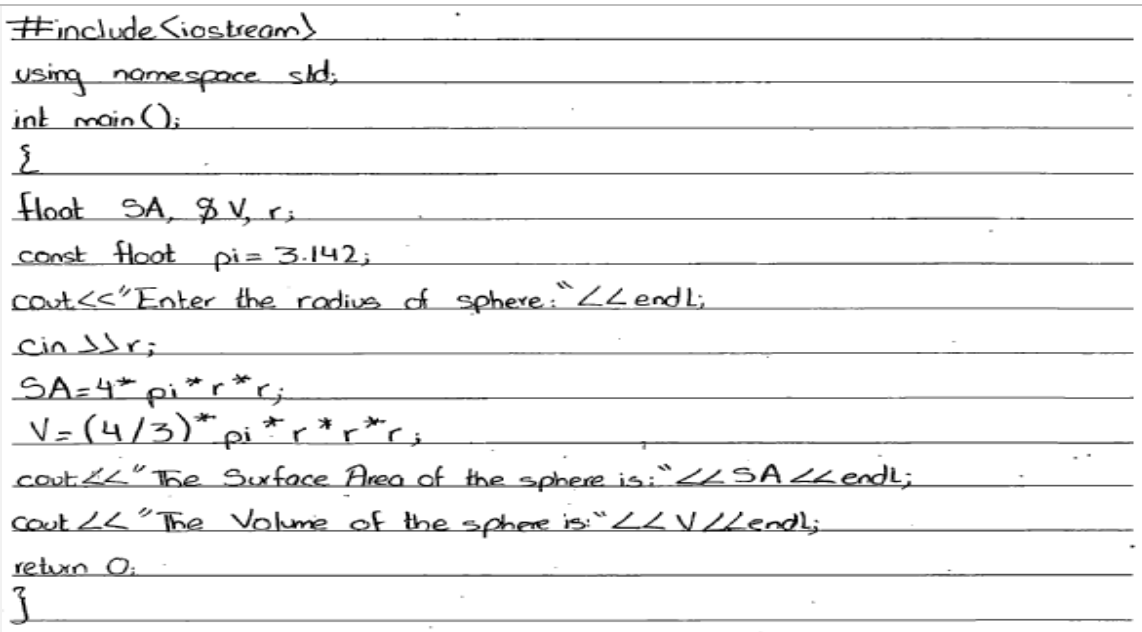
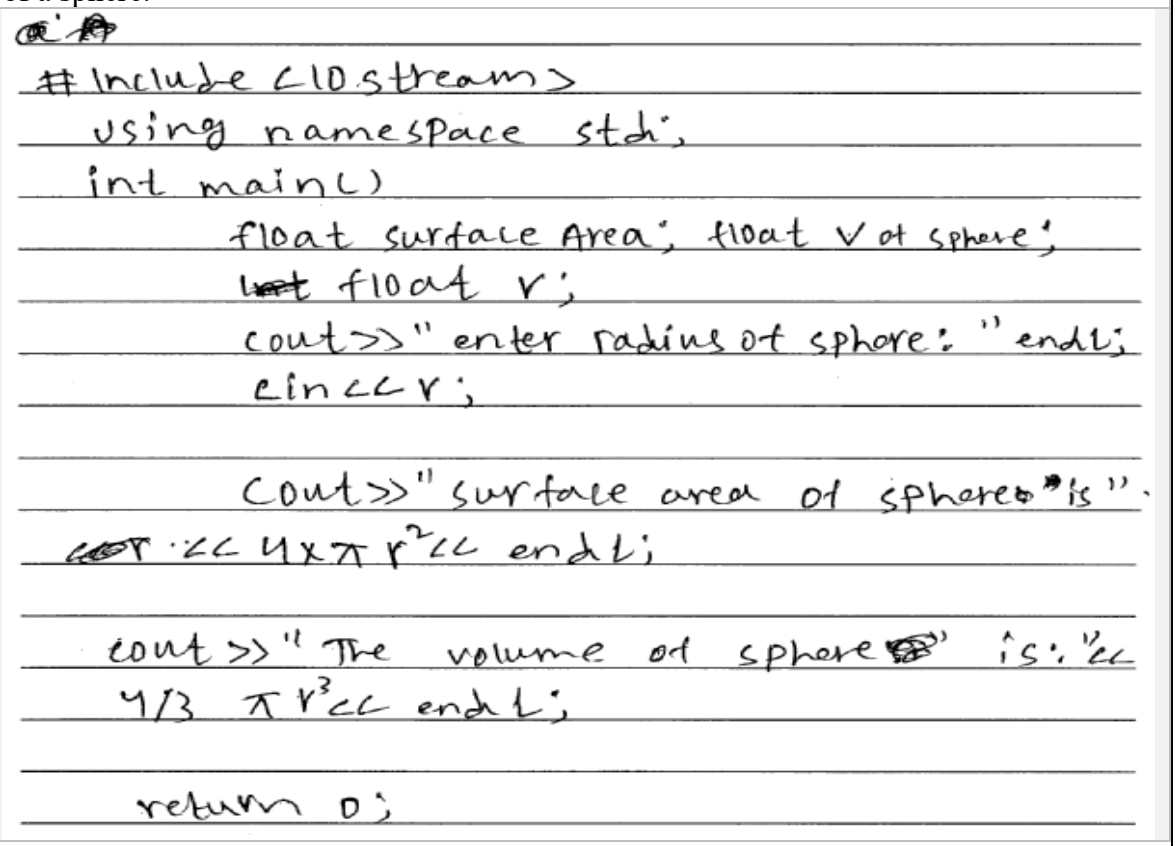
Maximising SLO Achievement	Preferred Pedagogy**Used for this SLO	Assessment Strategies
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Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Teachers are suggested to give candidates hands-on practice in writing, opening, and viewing text files from their local systems. Encourage experiments like writing user input to files or appending lines to existing files. Emphasise debugging and testing to validate each stage of file operation.		

Extended Response Questions (ERQs)

Extended response questions offered a choice between parts 'a' and 'b'

Question No. 7a	
Question Text	Write a C++ program for calculating and printing the surface area and volume of a sphere using a formula. Given is the formula to calculate the surface area and volume of a sphere. • Surface Area = 4πr2 • Volume of Sphere = 4/3πr3 (Note: The value of π = 3.142 is declared as a symbolic constant.)
SLO No.	11.4.3
SLO Text	Write a program using arithmetic operators.
Max Marks	7
Cognitive Level	Application
Checking Hints	1 mark for declaring PI as symbolic constant OR as variable 1 mark for declaring variable. 1 mark for getting input from the user. 1 mark for applying each correct formula (TWO required). 1 mark for displaying each correct output (TWO required).
Overall Performance	The majority of the candidates opted for this part of the question. But the overall performance of this question was low. Candidates struggled to correctly calculate and print the surface area and volume of a sphere using the given formula. Many candidates struggled to write the program completely. They missed the input, did not declare PI as a symbolic constant, and were not able to apply the formula.
Description of Better Responses	*In better responses,* candidates declared π (PI) as a symbolic constant with the value 3.142, took the radius as input, and correctly applied the formulas Surface Area = 4 × π × r^2 and Volume = (4/3) × π × r^3. They structured the program properly with header files, the main() function, variable declarations, and accurate use of constants, and then displayed the calculated surface area and volume clearly using cout, showing a sound understanding of formulas, input/output, and coding conventions in C++.

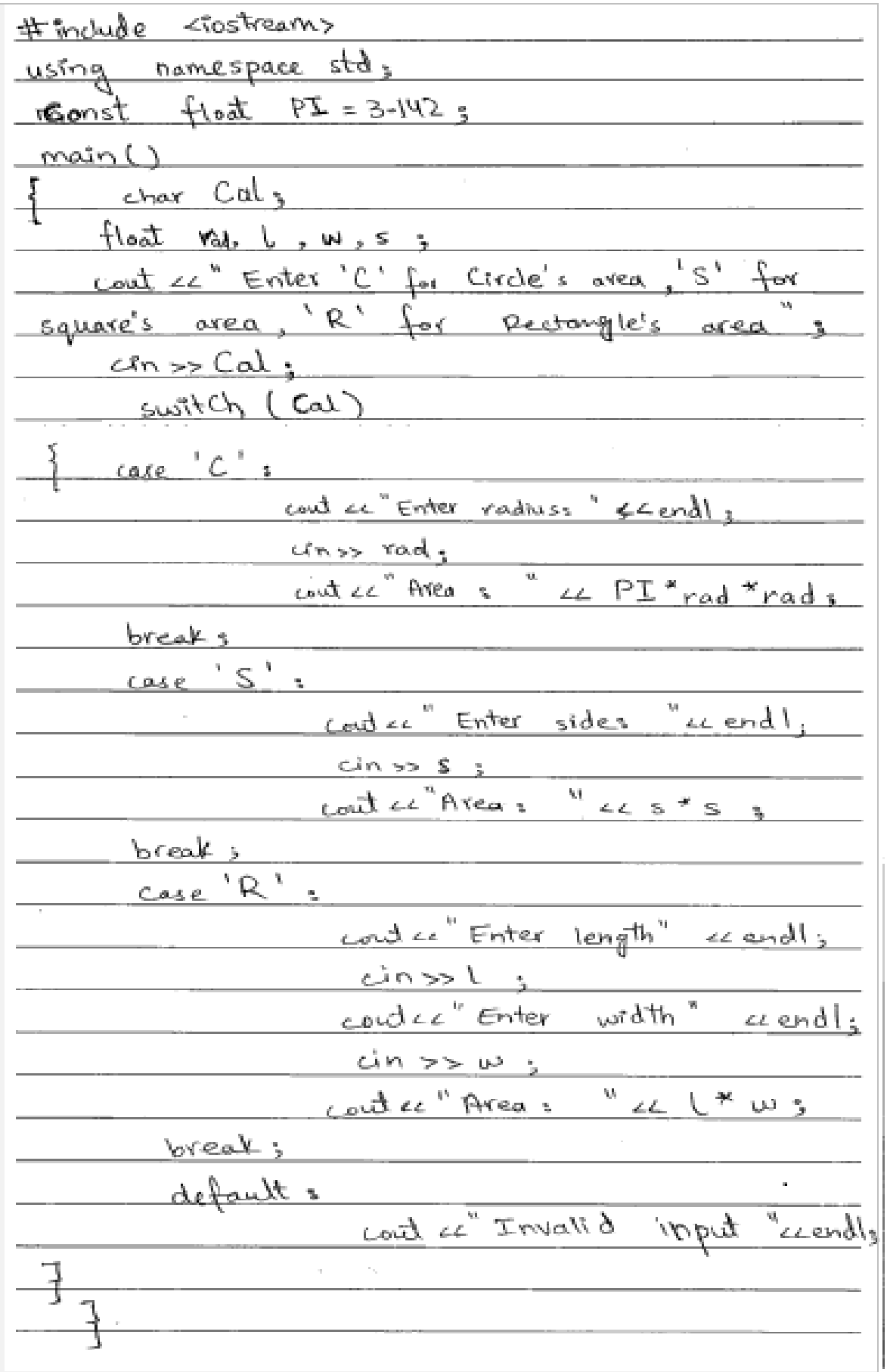
Image of Better Response	
Description of Weaker Responses	*In weaker responses,* candidates incorrectly took the radius as input from the user (some used the wrong data type or syntax), used PI directly as a value or did not use it at all, and applied the wrong formula, which led to incorrect results for the surface area and volume of a sphere.
Image of Weaker Response	

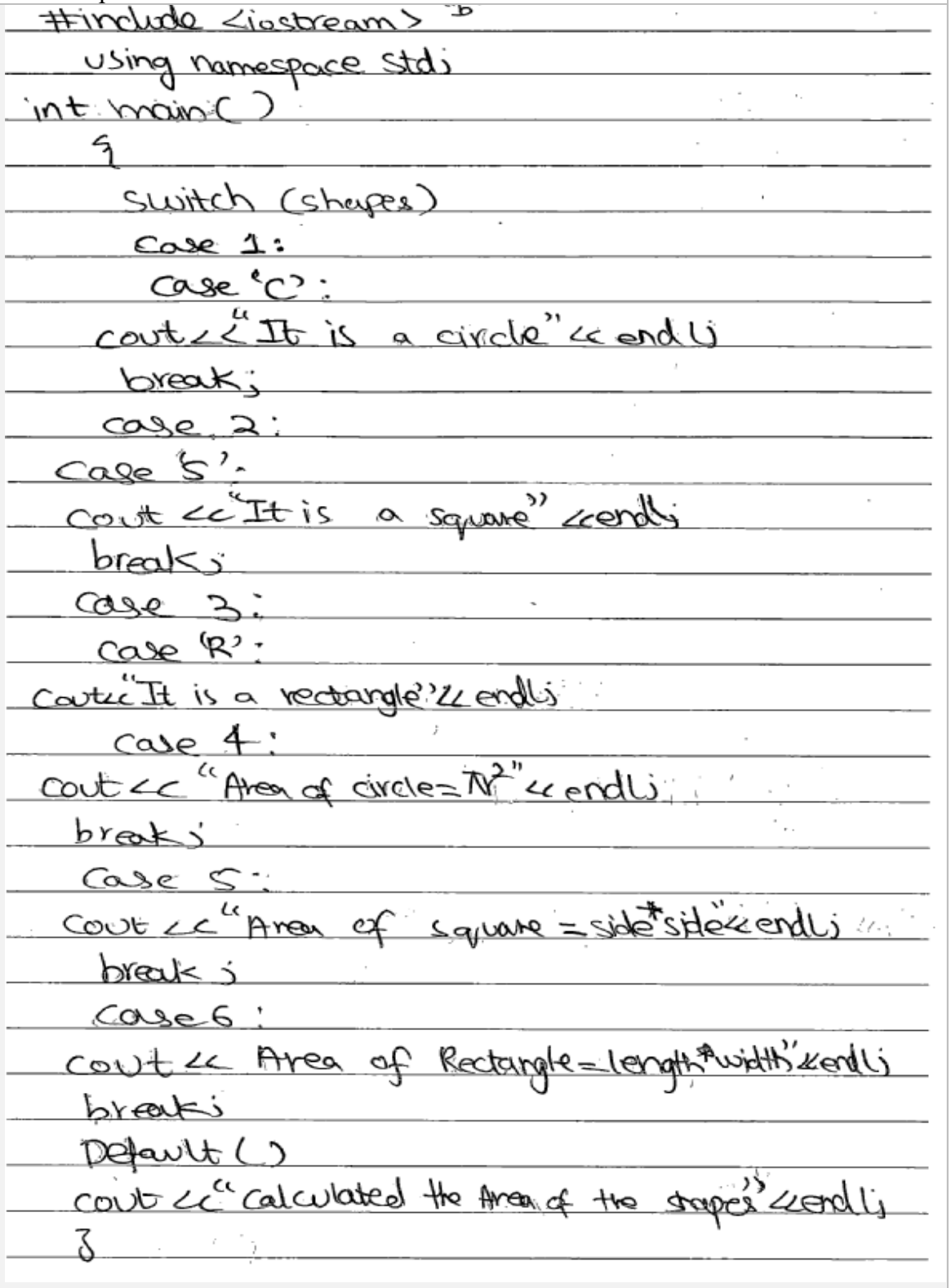
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy**Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform

cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Teachers are suggested to emphasise the correct use of data types and syntax in input statements, guide candidates to declare PI as a symbolic constant, and provided extra practice on applying the correct formulas. They had also reviewed common errors through targeted exercises to strengthen candidates' programming accuracy.		

Question No. 7b	
Question Text	Consider a simple geometric calculator that calculates the area of different shapes: circle, square, and rectangle. - program for this calculator using a switch statement that • Inputs a character representing the shape ('C' for circle, 'S' for square, 'R' for rectangle). • Uses the given formula to calculate the area of a circle, a square, and a rectangle. • Circle Area = πr2 • Square Area = side * side • Rectangle Area = length * width • inputs the necessary dimensions as float data type for each shape. • outputs the calculated area of the shape. (Note: The value of π = 3.142 is declared as a symbolic constant.)
SLO No.	12.1.5
SLO Text	Write a program using a switch statement.
Max Marks	7
Cognitive Level	Application
Checking Hints	1 mark for declaring PI as a symbolic constant. 1 mark for declaring a variable. 1 mark for getting input from the user. 3 marks for applying switch structure (THREE cases required). 1 mark for applying the 'break' statement in all cases.
Overall Performance	Though it was a choice question yet most of the candidates who opted for it did not attempted it well. Candidates struggled to correctly declare the variables, take input from the user, and apply the formula using a switch statement. Some candidates even showed a misconception regarding the switch statement as they were not familiar with the structure, applied the wrong syntax, and did not follow the proper program structure.
Description of Better Responses	*In better responses,* candidates wrote a well-structured C++ program that declared π (PI) as a symbolic constant with the value 3.142 and used float data types for the required dimensions. They correctly took input for the chosen shape ('C' for circle, 'S' for square, and 'R' for rectangle) and applied a switch statement with appropriate case labels. For each case, they implemented the formula accurately: πr^2 for a circle, side × side for a square, and length × width for a rectangle. Break statements were used properly to avoid fall-through, and a default case was included to handle invalid input. Finally, the calculated area of the selected shape was displayed clearly using cout, reflecting a good understanding of control structures, constants, input handling, and program syntax in C++.

Image of Better Response	
Description of Weaker Responses	*In weaker responses*, candidates declared variables incorrectly or skipped them, did not take proper input, and applied the switch statement with errors. They used wrong or missing case labels, applied incorrect syntax, omitted break statements, or failed to include the default case. Their programs lacked proper structure and produced incorrect

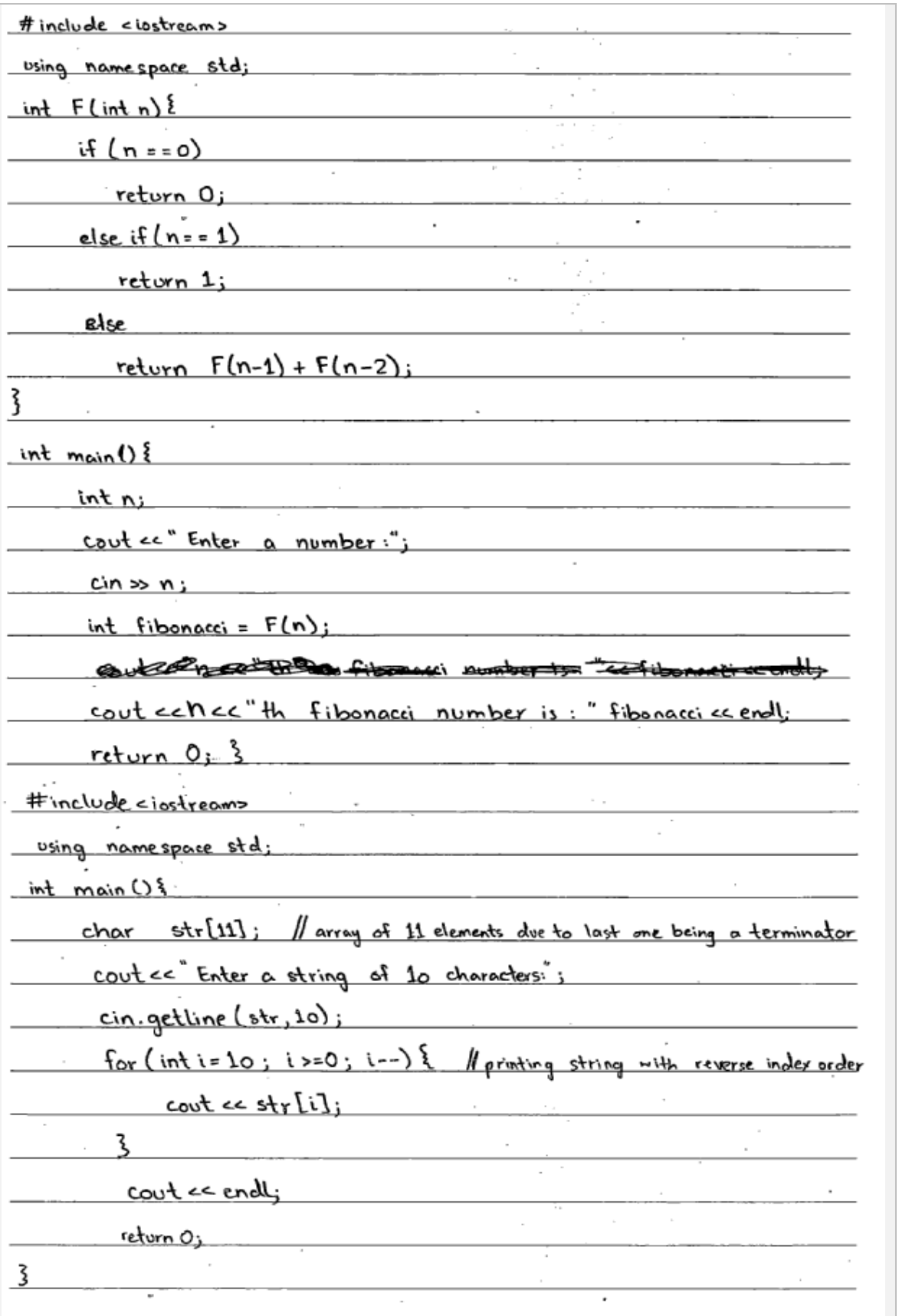
	or no output.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy**Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform

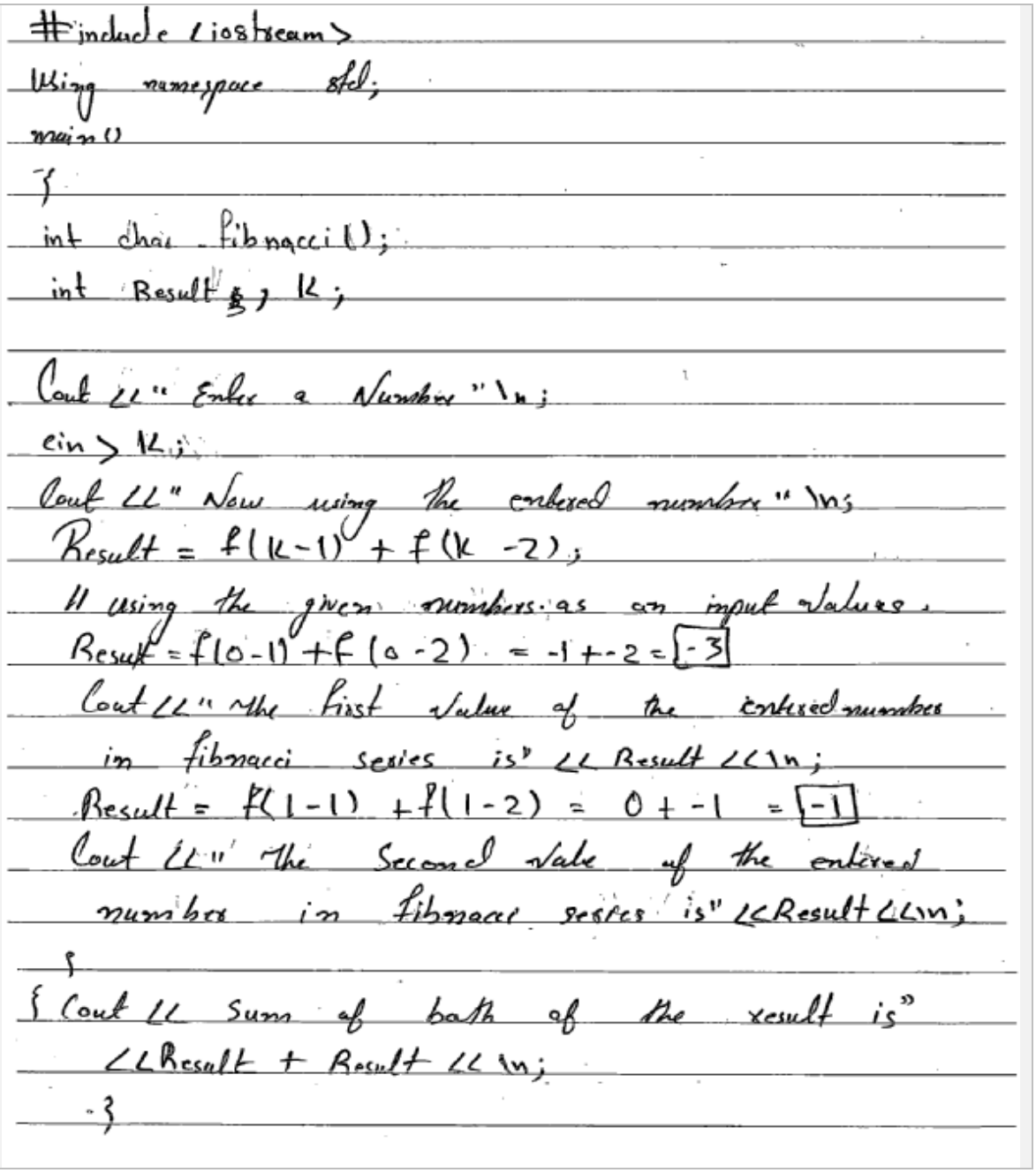
taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Teachers are suggested to reinforce the correct structure of the switch statement by giving targeted exercises on case labels, break statements, and the default case. They had provided practice tasks that compared switch with if-else to highlight differences and guided candidates through debugging activities to correct syntax and logical errors.		

Question No. 8a	
Question Text	Write a program using a user-defined function named 'fibonacci' that would generate the Fibonacci series in which each term is the sum of the previous two terms till the end of the selected range, and invoke it in the main function. • The formula of the Fibonacci series is: $F(n) = F(n-1) + F(n-2)$ • With the initial values: $F(0) = 0$, $F(1) = 1$
SLO No.	14.2.1
SLO Text	Write a program to invoke a user-defined function and pass arguments by constant, value and reference.
Max Marks	7
Cognitive Level	Application
Checking Hints	1 mark for declaring, initialising variables, and getting input for the range. 1 mark for loop structure. 1 mark for writing each correct logic of the function (FOUR statements required). 1 mark for invoking the function in main( ).
Overall Performance	Many of the candidates did not opt for this question, and the ones who opted even did not perform well at all. Candidates struggled with it; some did not know the concept of the Fibonacci series, even though the steps for generating the series were provided in the question. Only a few candidates managed to attempt it successfully.
Description of Better Responses	*In better responses,* candidates wrote a C++ program that defined a user-defined function named fibonacci to generate the Fibonacci series, where each term was calculated as the sum of the previous two terms using the formula $F(n) = F(n-1) + F(n-2)$ with initial values $F(0) = 0$ and $F(1) = 1$. They correctly declared the necessary variables, took the range as input from the user, and used a loop within the function to generate and print the series up to the specified range. The function was invoked properly in the main() function, and the output clearly displayed the Fibonacci series, showing a good understanding of functions, loops, and sequence generation in C++.

Image of Better Response	
	#include <iostream>
	using namespace std;
	int F(int n){
	if (n==0)
	return 0;
	else if(n==1)
	return 1;
	else
	return F(n-1) + F(n-2);
	}
	int main(){
	int n;
	cout << " Enter a number:";
	cin >> n;
	int fibonacci = F(n);
	~~cout << "th fibonacci number is: " << fibonacci << endl;~~
	cout <<n<< "th fibonacci number is : " fibonacci << endl;
	return 0; }
	#include <iostream>
	using namespace std;
	int main(){
	char str[11]; // array of 11 elements due to last one being a terminator
	cout<< " Enter a string of 10 characters:";
	cin.getline(str,10);
	for(int i=10; i>=0; i--){ // printing string with reverse index order
	cout << str[i];
	}
	cout << endl;
	return 0;
	}

Description of Weaker Responses	*In weaker responses*, candidates showed confusion about the logic of the Fibonacci series. Some did not initialise the first two terms correctly, while others skipped the loop or applied it with the wrong condition. Many candidates used incorrect formulas, such as multiplying or subtracting the terms instead of adding them, which led to invalid outputs. A very few candidates attempted to hardcode the series instead of generating it programmatically, and several responses contained syntax errors that prevented execution.

Image of Weaker Response



```
#include <iostream>
Using namespace std;
main()
{
int char fibnacci();
int Result, k;

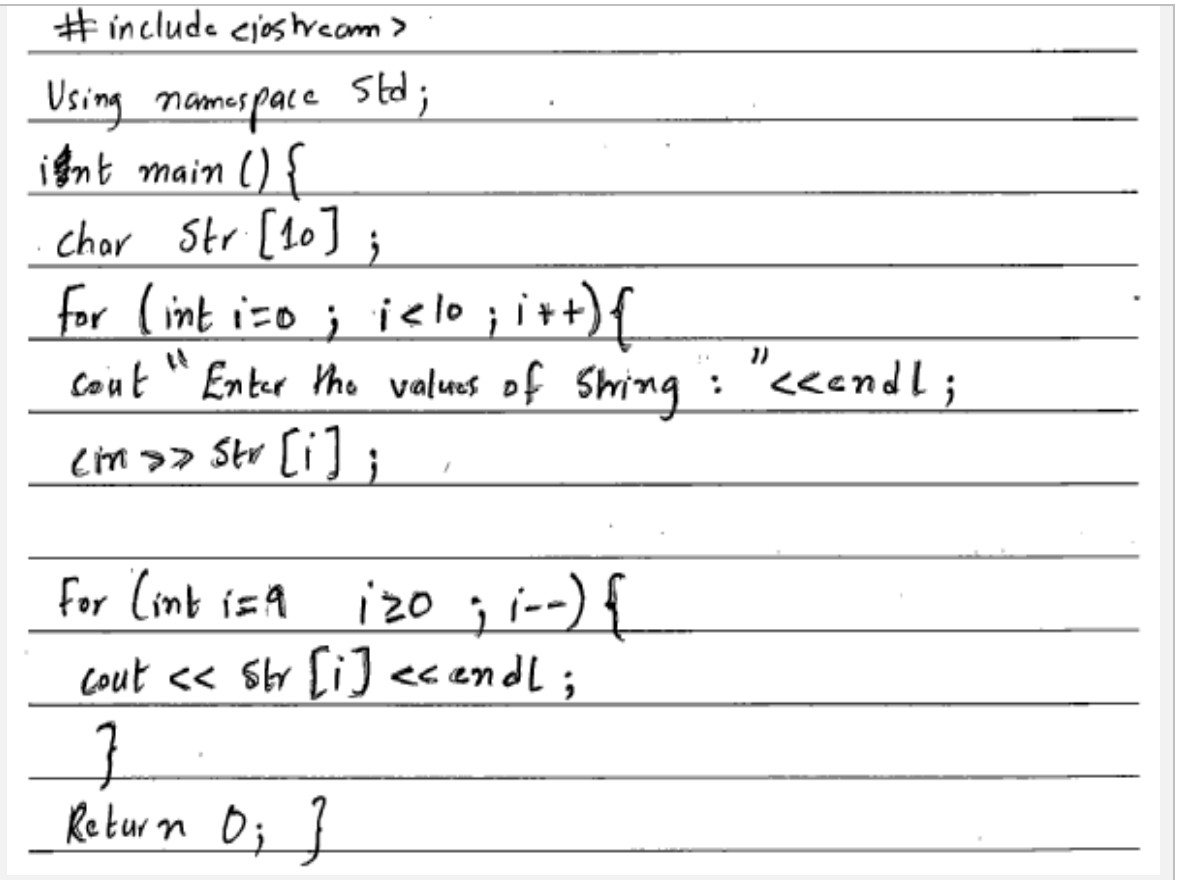
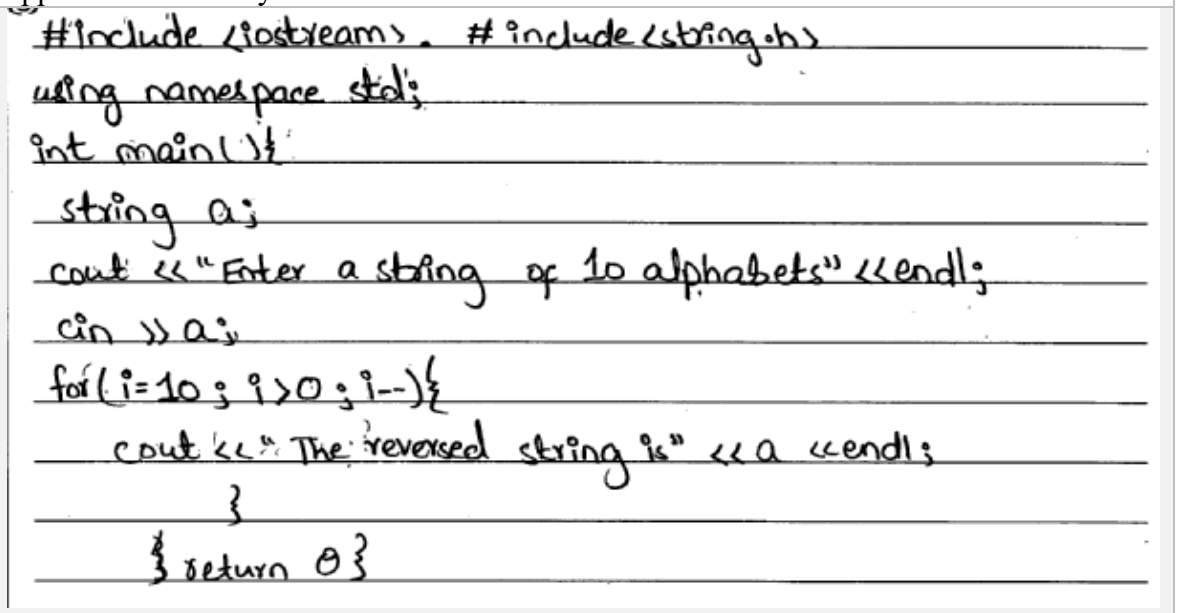
Cout << " Enter a Number " \n;
cin > k;
Cout << " Now using the entered number " \n;
Result = f(k-1) + f(k -2);
// using the given numbers as an input values.
Result = f(0-1) + f(0-2) = -1 + -2 = -3
Cout << " The first value of the entered number
in fibnacci series is" << Result << \n;
Result = f(1-1) + f(1-2) = 0 + -1 = -1
Cout << " The Second vale of the entered
number in fibnacci series is" << Result << \n;
}
{ Cout << Sum of both of the result is"
<< Result + Result << \n;
}
```

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy**Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Teachers are suggested to provide more practice on initialising variables, applying the correct formula, and using loops to generate the Fibonacci series or any other series, using practical classes.

Question No. 8b	
Question Text	Write a C++ program that reverses an inputted string of 10 alphabets and prints it on the screen.
SLO No.	13.1.7
SLO Text	Write a program which stores numeric values in a one-dimensional array using for loop and finds the highest, lowest and average values.
Max Marks	7
Cognitive Level	Application
Checking Hints	1 mark for declaring variable. 1 mark for taking input. 1 mark for using for loop (TWO required). 1 mark for including header file. 1 mark for prompting msg for input. 1 mark for writing the reversed string.
Overall Performance	The overall performance of this question was way below average. Only a small proportion of candidates achieved full marks, while a comparable number scored zero. Most candidates demonstrated partial understanding, with responses ranging between correct logic and significant errors in implementation.
Description of Better Responses	*In better responses*, candidates correctly declared a character array of fixed length, accepted the input of ten alphabets, and applied the proper logic to reverse the string. They used correct loop control to swap characIn better responses, candidates wrote a C++ program that accepted a string of 10 alphabets as input from the user, stored it in a character array or string variable, and then used a loop or built-in function to reverse the order of characters. They correctly applied logic by starting from the last character and swapping or printing sequentially towards the first, ensuring the string was reversed accurately. Finally, the reversed string was displayed on the screen using cout, demonstrating clear understanding of string handling, loops, and input/output in C++.ters from both ends, handled the index positions accurately, and displayed the reversed string on the screen without errors. Some candidates also adopted an alternative method by using built-in string functions to reverse the input.

Image of Better Response	
Description of Weaker Responses	*In weaker responses*, candidates did not handle the input correctly, either by taking fewer characters or including spaces. Some attempted the reversal using loops but applied wrong index positions, skipped swapping, or used incorrect loop conditions, which led to errors. Others who tried the built-in string functions either used the wrong function or applied it incorrectly.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy**Used for this SLO	Assessment Strategies
• Identify the expectation of	• Story Board • Cause and Effect	• Past paper questions • Discussion on E-Marking Notes

command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Teachers are suggested to provide more practice on string handling through character arrays and loop-based reversal, and to demonstrate the correct use of built-in string functions to strengthen candidates' understanding through multimedia classes.		

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.

Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgments

The Aga Khan University Examination Board (AKU-EB) acknowledges with gratitude the invaluable contributions of all the dedicated individuals who have played a pivotal role in the development of the Computer Science HSSC-II E-Marking Notes.

We extend our sincere appreciation to Syed Muhammad Waqas, Specialist Computer Science at AKU-EB, for taking the subject lead during the entire process of e-marking.

We particularly thank Ms. Sumaira Farhan, PECHS Girls School, Karachi, for evaluating each question's performance, delineating strengths and weaknesses in candidates' responses, and highlighting instructional approaches along with recommendations for better performance.

Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies, along with suggestions for improvement.

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• Munira Muhammad, Manager, Assessment, AKU-EB

• Zain Muluk, Manager, Examination Development, AKU-EB

• Dr. Naveed Yousuf, CEO, AKU-EB